

BIOINFORMATICS

Week 2 – Day 8

Mini Project 1: End-to-End Analysis of a Gene

Step 1 – Chosen Gene For this project : INS (Human Insulin)

Step 2 – Created a Project Folder in GitHub

Named it - Mini_Project_01_Insulin/

Made sub folders :

```
|— Data/  
|— BLAST/  
|— Alignment/  
|— Phylogeny/  
|— Structure/  
|— Python/  
|— Images/  
|— Report/  
|— README.md
```

Step 3 – Defined the Objective and created README.md

Step 4 – Retrieved the Data

Downloaded:

- DNA - Human insulin DNA (FASTA)
- Protein - Human insulin protein (FASTA)

Also downloaded insulin protein sequences from: Chimpanzee, Mouse, Cow, Zebrafish

Saved them inside: Data/

Step 5 – BLAST Analysis

Run: BLASTn

Run: BLASTp

Save in : BLAST/

Step 6 – Multiple Sequence Alignment

Aligned the five insulin proteins.

Saved : Alignment file and Screenshot.

Step 7 – Phylogenetic Tree

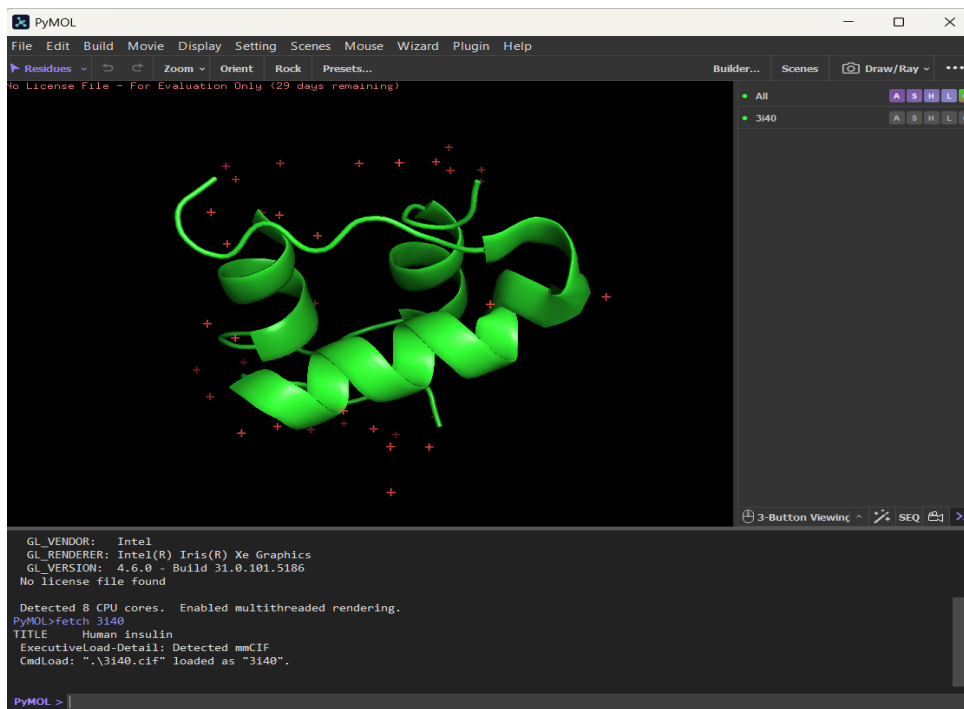
Used the alignment to build a tree and saved the image.

Step 8 – Protein Structure

PyMOL Commands to Open insulin in PyMOL.

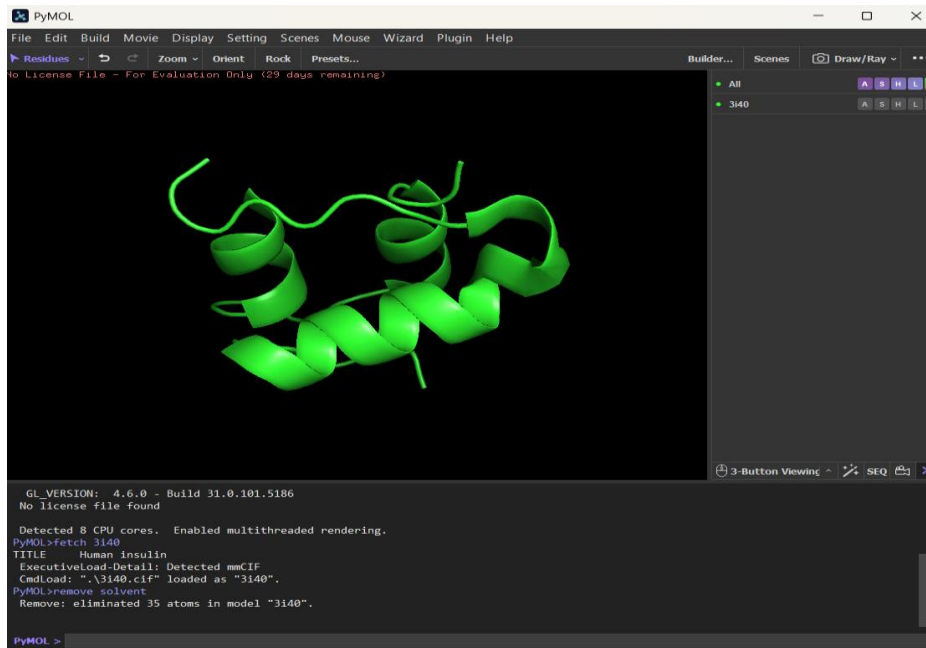
1. Fetch the 3D atomic coordinates directly from the PDB registry

fetch 3i40



2. Clean up the scene by removing crystallographic water molecules

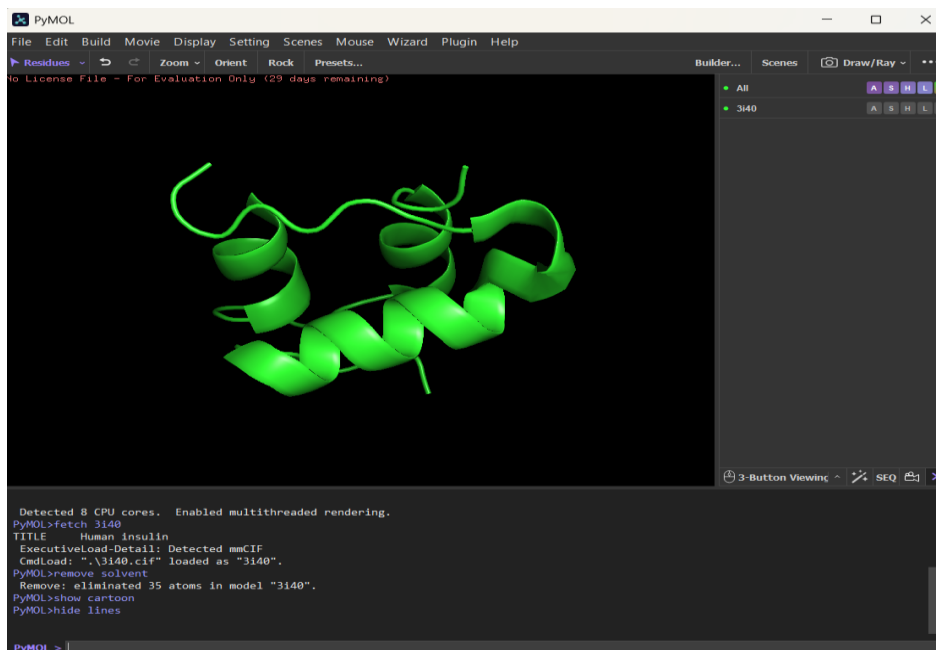
remove solvent



3. Apply the standard presentation layout

show cartoon

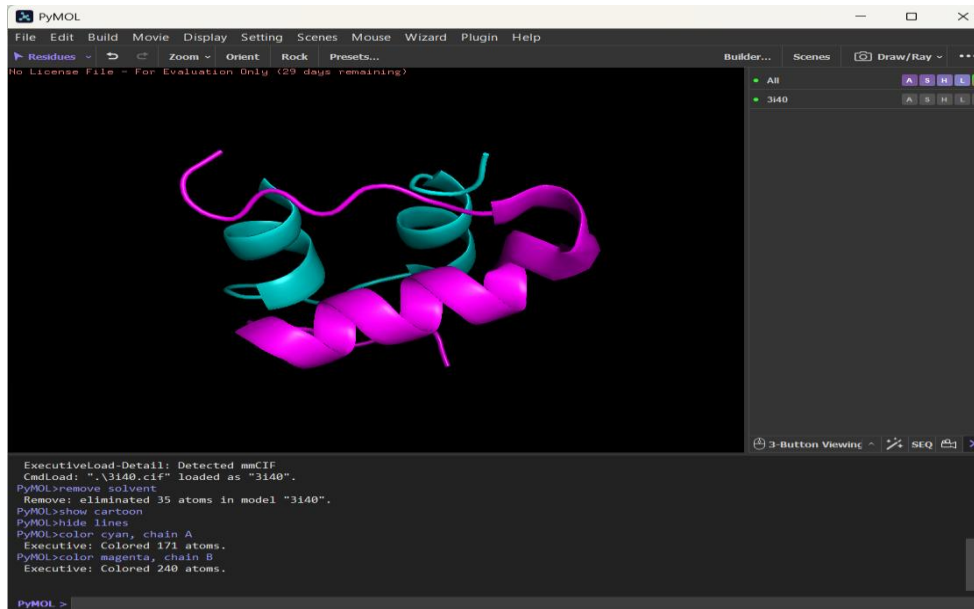
hide lines



4. Color the distinct polypeptide chains to see the boundaries clearly

color cyan, chain A

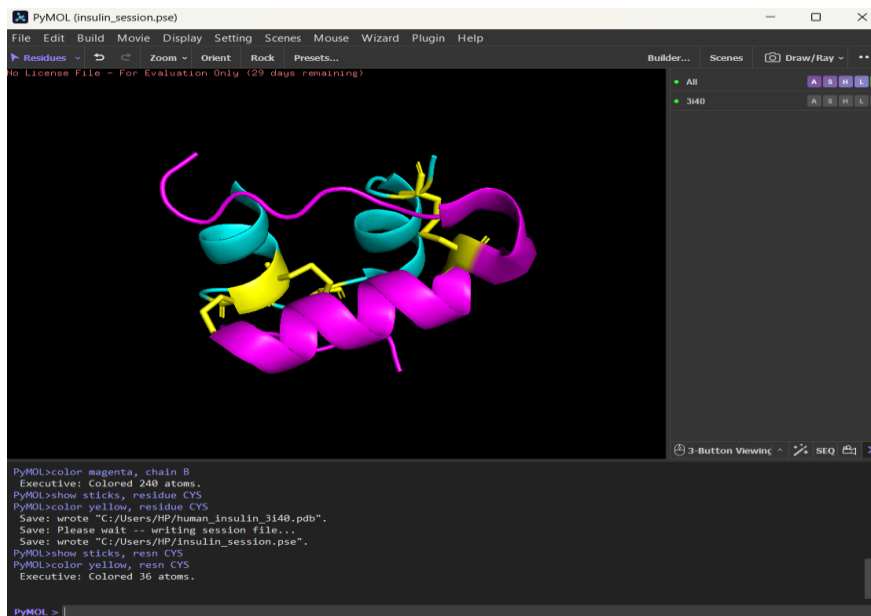
color magenta, chain B



5. Locate and visually highlight the specific disulfide bridge links

show sticks, residue CYS

color yellow, residue CYS



Step 9 – Python Component

Wrote one useful Python program that contributes to the project.

(GC Content Calculator combined with Nucleotide Counter)

Step 10 – Write the Report